

Chapter 4

Linear Programming

In this section, we consider the celebrated algorithm for computing an optimal solution in a linear program. The bedrock of this chapter is the collection of Theorem 3.22, 3.23, and 3.24. In particular, this method is defined only for polyhedra defined on a one- or two-sided bounded polyhedron.

4.1 Fundamental Theorem of Linear Programming

We take the following theorem as given. Let $f : X \rightarrow \mathbb{R}$ and $X^* = \{x^* \in X : f(x^*) = \inf_{x \in X} f(x)\}$. Observe that if f is not bounded from below, then $X^* = \emptyset$; for its contrapositive, if $X^* \neq \emptyset$, then f is bounded from below.

Proposition 4.1. *Let f be a convex function over X and $X \subseteq \mathbb{R}^n$ such that $X \neq \emptyset$. Let $X^* \subseteq X$ be the set whose elements $x \in X^*$ attain the minimum over X . If $X^* \cap \text{ri}(X) \neq \emptyset$, then f is constant over X ; that is, $X = X^*$.*

Theorem 4.2 (Bauer's Extreme Point Theorem for Convex Maximization). *Let f be a convex function defined on a closed convex set $C \subseteq \mathbb{R}^n$, where C contains one or more extreme points. Let $C^* \subseteq C$ be the set of elements that attain the maximum of f . If $C^* \neq \emptyset$, then C^* contains one or more extreme points of C .*

Proof: We proceed with induction on the dimension of C . Suppose that $\dim(C) = 0$, then C contains only a single point, which is an extreme point. Meanwhile, the only point in C naturally attains the maximum of f .

Inductively, suppose that $\dim(C) = r > 0$ and that every convex function f_0 defined on a convex set $C_0 \subseteq \mathbb{R}^n$, with dimension $r - 1$ or less and contains one or more extreme points, has that, if $C_0^* \neq \emptyset$, then C_0^* contains one or more extreme points of C_0 , where C_0^* is the set of points in C_0 that attain the maximum of f_0 over C_0 .

Let $\mathbf{x}^* \in C^*$. By Proposition 4.1, if $\mathbf{x}^* \in \text{ri}(C)$, then f is constant over C and, thus, $C^* = C$ and we are done since C contains one or more extreme point. Otherwise, we have that $\mathbf{x}^* \in C \setminus \text{ri}(C)$. By Theorem 3.30 and Proposition 1.11 (i.e., the proper separation theorem and that convex sets have nonempty relative interior), there is a hyperplane H that separates $\mathbf{x}^*$ from $\text{ri}(C)$; since $\mathbf{x}^* \in C$, we also have $\mathbf{x}^* \in C \cap H$. By Lemma 3.27, we have that $\dim(C \cap H) < \dim(C)$ and $f|_{C \cap H}$ remains convex. Apply Theorem 3.31 and $C \cap H \subseteq C$ and C does not contain a line subspace and, thus, neither does $C \cap H$ contain a line; therefore, $C \cap H$ contains one or more extreme points by Theorem 3.31 again.

Inductively, let C_1^* be the set of elements that attain the maximum of $f|_{C \cap H}$; since $\mathbf{x}^* \in C^*$, we have $\mathbf{x}^* \in C_1^*$, which implies that elements in C_1^* attain the minimum of f over C , and $C_1^* \subseteq C^*$. By induction, as $\dim(C \cap H) < \dim(C)$, we have that C_1^* contains one or more extreme points of $C \cap H$. By Theorem 3.28, which states that the extreme point in $C \cap H$ is an extreme point in C , we have that C^* contains one or more extreme points in C , which was what we wanted. $\square$

Theorem 4.3 (Fundamental Theorem of Linear Programming). *Given $\mathbf{c} \in \mathbb{R}^n$ and a polyhedral set $P \subseteq \mathbb{R}^n$ with one or more extreme points, consider a linear function $\langle \mathbf{c}, \cdot \rangle$ defined over P . Either f is not bounded from above, or there is an extreme point in P that attains the maximum $\sup_{\mathbf{x} \in P} f(\mathbf{x})$.* $\square$

4.2 Extrinsic Characterization for Extreme Points

Given a polyhedral set P , a point $\mathbf{x} \in \mathbb{R}^n$ is called a *vertex in P* if $\mathbf{x} \in P$ and there is some $\mathbf{c} \in \mathbb{R}^n$ so that $\langle \mathbf{c}, \mathbf{x} \rangle > \langle \mathbf{c}, \mathbf{x}' \rangle$ for all $\mathbf{x}' \in P \setminus \{\mathbf{x}\}$. It turns out that the definitions of a vertex, an extreme point, and a basic feasible solution are the same object viewed from three different perspectives. Before unifying the three characterizations of a polyhedral corner, it is worth reflecting on the logical architecture of our exposition. In Chapter 3, we established the fundamental equivalence between extreme points and basic feasible solutions (extreme point $\iff$ basic feasible solution). This ordering is both deliberate and necessary: extreme points (geometric/topological) and basic feasible solutions (algebraic/algorithmic) are intrinsic to the polyhedron P itself, depending solely on its defining constraint system without reference to an objective function. By isolating this intrinsic geometry-to-algebra bridge first in Theorem 3.22, we established a constructive computational skeleton for P . In this section, we introduce the extrinsic optimization perspective: the notion of a vertex defined relative to an optimization objective $\langle \mathbf{c}, \cdot \rangle$; and we close the loop by synthesizing all three viewpoints into a unified equivalence.

Theorem 4.4 (The Unification Theorem). *For a polyhedral set $P \subseteq \mathbb{R}^n$ and a point $\mathbf{x}^* \in \mathbb{R}_+^n$. The following are equivalent.*

1. The point $\mathbf{x}^*$ is a vertex in P .
2. The point $\mathbf{x}^*$ is an extreme point in P .
3. The point $\mathbf{x}^*$ is a basic feasible solution to P .

Proof: Theorem 3.22 shows that the point is an extreme point if and only if it is a basic feasible solution to P . We first show that a point $\mathbf{x}^*$ is a vertex in P implies that it is an extreme point in P ; next, we show that a point is a basic feasible solution to P implies that it is a vertex in P .

Suppose that $\mathbf{x}^* \in P$ is a vertex. By definition, there exists $\mathbf{c} \in \mathbb{R}^n$ such that for all $\mathbf{x} \in P \setminus \{\mathbf{x}^*\}$, we have that $\langle \mathbf{c}, \mathbf{x}^* \rangle > \langle \mathbf{c}, \mathbf{x} \rangle$. This says, for any two other points $\mathbf{u}, \mathbf{w} \in P$ and any $\alpha \in (0, 1)$, we have that

$$\langle \mathbf{c}, \mathbf{x}^* \rangle > \alpha \langle \mathbf{c}, \mathbf{u} \rangle + (1 - \alpha) \langle \mathbf{c}, \mathbf{w} \rangle.$$

Thus, there are no lines with endpoints in P that contain $\mathbf{x}^*$.

Suppose that $\mathbf{x}^* \in P$ is a basic feasible solution to P . Without loss of generality, we may assume $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r$ are linear independent active constraints so that $\langle \mathbf{a}_j, \mathbf{x}^* \rangle = b_j$ that attains the bounds b_j , for $j \in \{1, 2, \dots, r\}$. We define a vector $\mathbf{c} = \sum_{i=1}^m \mathbf{a}_i$. Since the constraints are active, we have that $\langle \mathbf{c}, \mathbf{x}^* \rangle = \sum_{i=1}^m b_i$. By the linear independence, $\mathbf{x}^*$ is the unique solution so that $\langle \mathbf{a}_j, \mathbf{x}^* \rangle = b_j$ for all $j \in \{1, 2, \dots, r\}$ and, for any other $\mathbf{x} \in P \setminus \{\mathbf{x}^*\}$, we have that $\langle \mathbf{a}_j, \mathbf{x}^* \rangle > \langle \mathbf{a}_j, \mathbf{x} \rangle$; this follows the definition of $\mathbf{x}^*$ is a vertex in P . $\square$

From Theorem 4.4, an extreme point defined geometrically now has an intrinsic and linear-algebraic characterization as a basic feasible solution, and an extrinsic definition as a vertex. Theorem 3.31 provided a geometric characterization of the existence of an extreme point; the next theorem provides a linear-algebraic characterization of the existence of an extreme point.

Theorem 4.5. Let $P \subseteq \mathbb{R}^n$ be a polyhedral set defined with m constraints and bounds $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m \in \mathbb{R}^n$ and $b_1, b_2, \dots, b_m \in \mathbb{R}$.

1. The polyhedral set P has one or more extreme point;
2. The polyhedral set P contains no line;
3. There is a list of n linearly independent constraints.

Proof: As Theorem 3.31 showed $(1) \iff (2)$, it suffices to show that $(1) \implies (3)$ and $(3) \implies (2)$. For $(1) \implies (3)$, suppose that there is an extreme point $\mathbf{x}^* \in P$; by Theorem 4.4, it is also a basic feasible solution in P and, by definition, there are n linearly independent active constraints with respect to $\mathbf{x}^*$.

For $(3) \implies (2)$, without loss of generality, suppose $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ is a list of n linearly independent constraints. Toward a contradiction, suppose there is a line $\ell = \{\mathbf{x} + \alpha \mathbf{d} : \alpha \in \mathbb{R}\} \subseteq P$ for some $\mathbf{x}, \mathbf{d} \in \mathbb{R}^n$ and $\mathbf{d} \neq \mathbf{0}$. By the definition of the polyhedral set P , we

have

$$\langle \mathbf{a}_i, \mathbf{x} + \alpha \mathbf{d} \rangle = \langle \mathbf{a}_i, \mathbf{x} \rangle + \alpha \langle \mathbf{a}_i, \mathbf{d} \rangle \leq b_i \quad \text{for } i \in \{1, 2, \dots, n\} \text{ and } \alpha \in \mathbb{R},$$

which implies that $\langle \mathbf{a}_i, \mathbf{d} \rangle = 0$ since, otherwise, the inequalities is violated by choosing α positively large enough or negatively large enough. Since $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^n$ is a list of linearly independent vector and $\langle \mathbf{a}_i, \mathbf{d} \rangle = 0$ for all $i \in \{1, 2, \dots, n\}$, we have that $\mathbf{d} = 0$, which contradicts the assumption that ℓ is a line. $\square$

4.3 Optimal Conditions and Degeneracy

We now consider the “basic” part of a basic feasible solution. Recall that a point $\mathbf{x} \in \mathbb{R}^n$ is a basic solution to $P \subseteq \mathbb{R}^n$ if all the equality constraints are active and there are n active constraints at $\mathbf{x}$ in total. In this section, we provide more insights on viewing a linear program with basic- and basic feasible solutions. We first restate Theorem 3.23, which establishes the equivalence between an extreme point and a basic feasible solution; we adjust the index notation to align with the convention in this chapter.

Theorem 4.6. *Let P be a standard form polyhedral set defined by a constraint matrix $M \in \mathbb{R}^{m \times n}$, consisting of rows $\mathbf{a}_1^\top, \mathbf{a}_2^\top, \dots, \mathbf{a}_m^\top$, and a bound vector $\mathbf{b} \in \mathbb{R}^m$, where each of its entries is $b_1, b_2, \dots, b_m$. Suppose that*

$$\dim(\text{span}(\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m\})) = k < m$$

and, without loss of generality, $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ is a list of linearly independent vectors. Define $Q = \{\mathbf{x} \in \mathbb{R}_+^n : \forall i \in \{1, 2, \dots, k\}, \langle \mathbf{a}_i, \mathbf{x} \rangle = b_i\}$. We have that $Q \neq \emptyset$ or $\mathbf{b} \in \text{im}(M)$, then $P = Q$. In particular, if $P \neq \emptyset$, then $P = Q$. $\square$

Assumption 4.7. *Throughout the section, we consider the same linear program and notation as defined below. Let $\mathbf{c} \in \mathbb{R}^n$, $M \in \mathbb{R}^{m \times n}$, and $\mathbf{b} \in \mathbb{R}^m$, define the linear program*

$$\max_{\mathbf{x} \in P} \langle \mathbf{c}, \mathbf{x} \rangle \tag{LPS}$$

where $P = \{\mathbf{x} : M\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$ is the standard form polyhedral set defined by the system of equalities and inequalities. Let B be an index set of a list of m linearly independent columns in M and $N = \{1, 2, \dots, n\} \setminus B$. Denote M by its column form so that $M = [\mathbf{f}_1 \mid \mathbf{f}_2 \mid \dots \mid \mathbf{f}_n]$ and $M_B = [\mathbf{f}_{B(1)} \mid \mathbf{f}_{B(2)} \mid \dots \mid \mathbf{f}_{B(m)}]$ be the full-rank $m \times m$ -submatrix. We denote each entry of $\mathbf{c}$ as c_j for $j \in \{1, 2, \dots, n\}$.

Theorem 3.23 restated. Under the assumption and notation as in Assumption 4.7, a

vector $\mathbf{v} \in P$ is an extreme point of P if and only if the columns of M with indices in $\text{supp}(\mathbf{v})$ are linearly independent.

For a matrix $M \in \mathbb{R}^{m \times n}$ and an index set $B \subseteq \{1, 2, \dots, n\}$ of order m , i.e. $|B| = m$; for convenience, we denote the i -th index in B by $B(i)$, for $i \in \{1, 2, \dots, m\}$. With respect to B , if the list of $B(1)$ -, $B(2)$ -, $\dots$, $B(m)$ -th columns of M are linearly independent, then we call them as the *basic columns*. In terms of solving a system of linear equations $M\mathbf{x} = \mathbf{b}$, we call the set of $B(i)$ -th variables of $\mathbf{x}$ the *basic variables*; we call the columns in M and the variables not in B *nonbasic*. Generally, we denote $M_B \in \mathbb{R}^{m \times m}$ as the submatrix of M whose i -th column being the $B(i)$ -th column of M , for $i \in \{1, 2, \dots, m\}$.

Remark 4.8. In this section, we may assume B is an index *set*; even more, we may assume $B = \{1, 2, \dots, m\}$ without loss of generality. However, in the following sections, starting with Section 4.5, we need B to be a list so that every replacement is performed in place.

Suppose B corresponds to m linearly independent columns of $M \in \mathbb{R}^{m \times n}$ and, thus, M_B is a full-rank $m \times m$ -matrix. The sub-system of equations of $M\mathbf{x} = \mathbf{b}$ then has a unique solution $[\mathbf{x}]_B \in \mathbb{R}^m$ to

$$M_B[\mathbf{x}]_B = \mathbf{b} \quad \text{or, equivalently} \quad [\mathbf{x}]_B = M_B^{-1}\mathbf{b};$$

recall that $[\mathbf{x}]_B$ indicates the subvector extracted from entries of $\mathbf{x}$ with indices in B .

Consider a point $\mathbf{x}$ so that the entries with their indices in B are formed by $[\mathbf{x}]_B$ and 0 otherwise. Observe that, $\text{supp}(\mathbf{x}) \subseteq B$ since not all entries in $[\mathbf{x}]_B$ are non-zero. By Theorem 3.23 and 4.4, we have that $\mathbf{x} \in P$ is a basic solution. For a polyhedral set in standard form, being basic and being basic feasible are equivalent.

Remark 4.9. The above operation to obtain $\mathbf{x}$ is sometimes called the construction of a basic solution. Some textbooks refer to it as an “algorithm starting by **choosing** (or **picking**) m linearly independent columns.” However, the last author of this booklet felt that it is misleading since the verb “choosing” refers to an explicit algorithm that should be introduced beforehand. However, the “choosing” part is implicitly given in the later procedure, implied by theorems instead of being explicitly extracted from a matrix; thus, we assume the linearly independent vectors are given for now.

In what follows, we consider the “elbow room” of a basic solution in P . For convenience, let the j -th entry of a vector $\mathbf{x}$ be x_j ; we denote $[\mathbf{x}]_j = [\mathbf{x}]_{\{j\}}$.

Definition 4.10. Consider Linear Program (LPS) in Assumption 4.7. For each nonbasic index $k \in N$, define the k -th *basic direction* $\mathbf{d}^{(k)} \in \mathbb{R}^n$ with respect to B ,

with each of the j -th entry of $\mathbf{d}^{(k)}$, as the following: define

$$[\mathbf{d}^{(k)}]_B = -M_B^{-1} \mathbf{f}_k, \quad [\mathbf{d}^{(k)}]_{N \setminus \{k\}} = \mathbf{0}, \quad \text{and} \quad [\mathbf{d}^{(k)}]_k = 1.$$

Define j -th reduced cost as

$$\bar{c}_j = c_j - \langle [\mathbf{c}]_B, M_B^{-1} \mathbf{f}_j \rangle$$

and form the reduced cost $\bar{\mathbf{c}}$ by the collection of all the j -th reduced cost.

Using the notations that are consistent with those in Definition 4.10, we remark on a few points. As M_B has full rank, the matrix M_B is invertible; so $[\mathbf{d}^{(k)}]_B$ is well-defined. For $j \in B$ defined by $j = B(i)$ for some $i \in \{1, 2, \dots, m\}$, the indicator vector $\boldsymbol{\chi}_i$ has that $M_B \boldsymbol{\chi}_i = \mathbf{f}_{B(i)}$ and, equivalently, $\boldsymbol{\chi}_i = M_B^{-1} \mathbf{f}_{B(i)}$; this is a reason for that basis direction is not defined for indices in B . For $j \in B$, we then have $\bar{c}_j = c_j - \langle [\mathbf{c}]_B, \boldsymbol{\chi}_{B(i)} \rangle = c_j - c_j = 0$ since, again, that M_B is invertible and bijective. Otherwise, when $j \in N$, we have $M_B^{-1} \mathbf{f}_j = -[\mathbf{d}^{(j)}]_B$ is the j -th basic direction. That is, fix a basis index B , we have

$$\bar{c}_j = \begin{cases} c_j - \langle [\mathbf{c}]_B, M_B^{-1} \mathbf{f}_j \rangle & \text{if } j \in N, \\ 0 & \text{otherwise.} \end{cases}$$

Finally, consider the vectorization that $\bar{\mathbf{c}}$ can be written in the form of

$$\bar{\mathbf{c}} = \mathbf{c} - \left([\mathbf{c}]_B^\top M_B^{-1} M \right)^\top = \mathbf{c} - (M_B^{-1} M)^\top [\mathbf{c}]_B.$$

Lemma 4.11. *Consider Linear Program (LPS) in Assumption 4.7. For $k \in N$, let $\mathbf{d}^{(k)} \in \mathbb{R}^n$ be the k -th basic direction with respect to B . We have that $M \mathbf{d}^{(k)} = \mathbf{0}$.*

Proof: We denote B by its column form so that $M = [\mathbf{f}_1 \mid \mathbf{f}_2 \mid \dots \mid \mathbf{f}_n]$ and let d_j denote the j -th entry of $\mathbf{d}^{(k)}$, for $j \in \{1, 2, \dots, n\}$. Note that a matrix multiplication $M \mathbf{d}^{(k)} = \mathbf{0}$ can be reinterpreted as a linear combination. For each $k \in N$, we obtain that

$$\begin{aligned} M \mathbf{d}^{(k)} &= \sum_{j=1}^n d_j \mathbf{f}_j \\ &= \sum_{j \in B} d_j \mathbf{f}_j + \sum_{j \in N} d_j \mathbf{f}_j \\ &= M_B [\mathbf{d}^{(k)}]_B + d_k \mathbf{f}_k + \sum_{j \in N \setminus \{k\}} d_j \mathbf{f}_j. \end{aligned}$$

By the definition that B chooses a column basis of M , the matrix M_B is invertible and,

thus, $[\mathbf{d}^{(k)}]_B = -M_B^{-1}\mathbf{f}_k$ and $d_k = 1$ and $d_j = 0$ otherwise, we continue so that

$$\begin{aligned}
M\mathbf{d}^{(k)} &= M_B[\mathbf{d}^{(k)}]_B + d_k\mathbf{f}_k + \sum_{j \in N \setminus \{k\}} d_j\mathbf{f}_j \\
&= M_B(-M_B^{-1}\mathbf{f}_k) + 1 \cdot \mathbf{f}_k + \sum_{j \in N \setminus \{k\}} 0 \cdot \mathbf{f}_j \\
&= -\mathbf{f}_k + \mathbf{f}_k \\
&= \mathbf{0}.
\end{aligned}$$

This finishes the proof. $\square$

Theorem 4.12. *Consider Linear Program (LPS) in Assumption 4.7 and let $\mathbf{x}^* \in \mathbb{R}^n$ be a basic feasible solution in P . Suppose that B also satisfies that $\text{supp}(\mathbf{x}^*) \subseteq B$ and $[\mathbf{x}^*]_N = \mathbf{0}$. If the reduced cost $\bar{\mathbf{c}}$ associated with B satisfies that $\bar{\mathbf{c}} \leq \mathbf{0}$, then $\mathbf{x}^*$ attains the maximum.*

Proof: Let $\mathbf{x} = \mathbf{x}^*$. Suppose $\mathbf{y} \in P \setminus \{\mathbf{x}\}$ is another basic solution in P ; we have that $M\mathbf{x} = M\mathbf{y} = \mathbf{b}$ and $\mathbf{y} - \mathbf{x} \neq \mathbf{0}$; consider the directional vector $\mathbf{u} = \mathbf{y} - \mathbf{x}$ that has $M\mathbf{u} = \mathbf{0}$. Denote $M = [\mathbf{f}_1 \mid \mathbf{f}_2 \mid \cdots \mid \mathbf{f}_n]$ and u_j, x_j, y_j denotes the j -th entries of $\mathbf{u}, \mathbf{x}, \mathbf{y}$, for $j \in \{1, 2, \dots, n\}$, to show that $\mathbf{x}$ is optimal, we show that

$$\langle \mathbf{c}, \mathbf{y} \rangle - \langle \mathbf{c}, \mathbf{x} \rangle = \langle \mathbf{c}, \mathbf{u} \rangle = \sum_{j=1}^n c_j u_j = \langle [\mathbf{c}]_B, [\mathbf{u}]_B \rangle + \sum_{j \in N} c_j u_j \leq 0$$

Consider the rearrangement $\mathbf{0} = M\mathbf{u} = M_B[\mathbf{u}]_B + \sum_{j \in N} u_j \mathbf{f}_j$, which gives that

$$[\mathbf{u}]_B = - \sum_{j \in N} u_j M_B^{-1} \mathbf{f}_j.$$

We then have that

$$\begin{aligned}
\langle \mathbf{c}, \mathbf{u} \rangle &= \langle [\mathbf{c}]_B, [\mathbf{u}]_B \rangle + \sum_{j \in N} c_j u_j \\
&= \left\langle [\mathbf{c}]_B, - \sum_{j \in N} u_j M_B^{-1} \mathbf{f}_j \right\rangle + \sum_{j \in N} c_j u_j \\
&= \sum_{j \in N} c_j u_j - \sum_{j \in N} u_j \langle [\mathbf{c}]_B, M_B^{-1} \mathbf{f}_j \rangle \\
&= \sum_{j \in N} (c_j - \langle [\mathbf{c}]_B, M_B^{-1} \mathbf{f}_j \rangle) u_j \\
&= \sum_{j \in N} \bar{c}_j u_j.
\end{aligned}$$

By the hypothesis that $[\mathbf{x}]_N = \mathbf{0}$ and $P \subseteq \mathbb{R}_+^n$, for $j \in N$, we have that $u_j = y_j - x_j = y_j - 0 = y_j \geq 0$. Since $\bar{\mathbf{c}} \leq 0$, we have that $\langle \bar{\mathbf{c}}, \mathbf{u} \rangle = \sum_{j \in N} \bar{c}_j u_j \leq 0$. $\square$

Let $P \subseteq \mathbb{R}_+^n$ be a polyhedral set defined by a matrix $M \in \mathbb{R}^{m \times n}$. Recall that a point $\mathbf{x}$ is a basic solution if there are n constraints that are active at $\mathbf{x}$. As the assumption is that $m \leq n$, if n constraints are active, then there are $n - m$ or more entries in $\mathbf{x}$ that attain the non-negative constraint, i.e., are zero. In other words, $\text{supp}(\mathbf{x}) \leq m$. A basic solution $\mathbf{x}$ is *degenerate* if there are more than n constraints active at $\mathbf{x}$.

Remark 4.13. Equivalently, a basic solution $\mathbf{x}$ is degenerate if $|\text{supp}(\mathbf{x})| < m$. In particular, a basic solution $\mathbf{x}$ is non-degenerate if $|\text{supp}(\mathbf{x})| = m$ and, by Theorem 3.23, $\text{supp}(\mathbf{x})$ corresponds to a list of m linearly independent columns in M .

Lemma 4.14. *The reduced cost of a non-degenerate basic solution in a polyhedral set is well-defined.*

Proof: Consider Linear Program (LPS) in Assumption 4.7 with a basic feasible solution $\mathbf{x}^*$. By Theorem 3.23, the indices in $\text{supp}(\mathbf{x}^*)$ correspond to linearly independent columns of M . Moreover, by the definition of degeneracy and Remark 4.13, the number of active constraints coincides with the cardinality of $|\text{supp}(\mathbf{x}^*)|$. Thus, under the definition of reduced cost, the full-rank submatrix induced from the standard form polyhedral set by the active constraints is unique, which implies well-definedness. $\square$

Lemma 4.15. *Consider Linear Program (LPS) in Assumption 4.7 and let $\mathbf{x}$ be a non-degenerate basic feasible solution so that $B = \text{supp}(\mathbf{x}^*)$. For every $k \in \{1, 2, \dots, n\} \setminus B$, there exists $\theta \in \mathbb{R}_+$ so that so that $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$, where $\mathbf{d}^{(k)}$ is the k -th basic direction.*

Proof: By Lemma 4.14, the set of basic indices $B = \text{supp}(\mathbf{x})$ and the invertible submatrix M_B are uniquely defined. Note that $\mathbf{d}^{(k)}$ is defined by $[\mathbf{d}^{(k)}]_B = M_B^{-1} \mathbf{f}_k$, $d_k = 1$, and $d_j = 0$ for $j \in N \setminus \{k\}$, where $\mathbf{f}_k$ is the k -th column of M and d_k and d_j are the entries of $\mathbf{d}^{(k)}$ with indices in N . Since we have $\mathbf{x}$ is non-degenerate, $[\mathbf{x}]_B > \mathbf{0}$ and $[\mathbf{x}]_N = \mathbf{0}$. We now have $\mathbf{x} + \theta \mathbf{d}^{(k)}$ is formed by

$$\begin{aligned} [\mathbf{x} + \theta \mathbf{d}^{(k)}]_B &= [\mathbf{x}]_B - \theta M_B^{-1} \mathbf{f}_k, \\ [\mathbf{x} + \theta \mathbf{d}^{(k)}]_k &= x_k + \theta d_k = x_k + \theta, \text{ and} \\ [\mathbf{x} + \theta \mathbf{d}^{(k)}]_{N \setminus \{k\}} &= [\mathbf{x}]_{N \setminus \{k\}}. \end{aligned}$$

By Lemma 4.11, we have $M(\mathbf{x} + \theta \mathbf{d}^{(k)}) = M\mathbf{x}$; thus, as long as we maintain $\mathbf{x} + \theta \mathbf{d}^{(k)} \geq \mathbf{0}$, we have that $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$. Since we have restricted $\theta > 0$, we need to control only the

entries with indices in B . If $\mathbf{d}^{(k)} \geq \mathbf{0}$, then we have $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$ for all $\theta \in \mathbb{R}_+$. Otherwise, we may assume there is one or more indices in B so that $[\mathbf{d}^{(k)}]_i < 0$. Let x_j and d_j denote the j -th entry of $\mathbf{x}$ and $\mathbf{d}^{(k)}$, respectively, for $j \in B$. Define

$$\theta^* = \min_{i \in B, [\mathbf{d}^{(k)}]_i < 0} \frac{[\mathbf{x}]_i}{-[\mathbf{d}^{(k)}]_i};$$

let $i^* \in B$ be the one that attains the minimum. By this choice, for every $j \in B$, we have that either

$$[\mathbf{x}]_j + \left(-\frac{[\mathbf{x}]_{i^*}}{[\mathbf{d}^{(k)}]_{i^*}} \right) [\mathbf{d}^{(k)}]_j \geq [\mathbf{x}]_j + \left(-\frac{[\mathbf{x}]_j}{[\mathbf{d}^{(k)}]_j} \right) [\mathbf{d}^{(k)}]_j = 0 \quad \text{if } [\mathbf{d}^{(k)}]_j < 0$$

or $[\mathbf{x}]_j + \theta [\mathbf{d}^{(k)}]_j \geq 0$ if $[\mathbf{d}^{(k)}]_j \geq 0$. This implies that $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$, which proves the statement. $\square$

Combining Lemma 4.15 and Theorem 4.12, a criterion for an objective function to be unbounded is as follows.

Corollary 4.16. *Consider Linear Program (LPS) in Assumption 4.7. If a basic feasible solution $\mathbf{x} \in P$ does not attain the maximum, so that $\bar{c}_k > 0$ and the basic direction $\mathbf{d}^{(k)} \geq \mathbf{0}$, then the objective function is unbounded from above.*

Proof: As in the statement, for some $k \in N$ so that $\bar{c}_k > 0$, by the construction as in Theorem 4.15, we have that $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$ for all $\theta \in \mathbb{R}_{++}$. By the construction as in Lemma 4.12, we have that $\langle \mathbf{c}, \mathbf{x} + \theta \mathbf{d}^{(k)} \rangle = \langle \mathbf{c}, \mathbf{x} \rangle + \theta \bar{c}_k$, which is arbitrarily large given the choice of θ . $\square$

Theorem 4.17. *Consider Linear Program (LPS) in Assumption 4.7. If $\mathbf{x}^*$ is a nondegenerate basic feasible solution and attains the maximum, then the reduced cost $\bar{\mathbf{c}}$ has that $\bar{\mathbf{c}} \leq \mathbf{0}$.*

Proof: Suppose $\mathbf{x}$ is a nondegenerate basic solution and is an optimal solution for (LPS). By Lemma 4.14, there is a unique index set B of cardinality m so that M_B is an invertible submatrix. Toward a contradiction, suppose that $\bar{\mathbf{c}} \not\leq \mathbf{0}$; that is, there is some $k \in \{1, 2, \dots, n\}$ so that $c_k > 0$.

Recall that $c_j = 0$ for all $j \in B$; we have that $k \in N$. By Lemma 4.15, let $\theta \in \mathbb{R}_{++}$ be an appropriate scalar so that $\mathbf{x}^* + \theta \mathbf{d}^{(k)} \in P$. Consider

$$\langle \mathbf{c}, \mathbf{x}^* + \theta \mathbf{d}^{(k)} \rangle = \langle \mathbf{c}, \mathbf{x}^* \rangle + \theta \langle \mathbf{c}, \mathbf{d}^{(k)} \rangle.$$

We now have

$$\langle \mathbf{c}, \mathbf{d}^{(k)} \rangle = \langle [\mathbf{c}]_B, [\mathbf{d}^{(k)}]_B \rangle + c_k = c_k - \langle [\mathbf{c}]_B, M_B^{-1} \mathbf{f}_k \rangle = \bar{c}_k$$

By the contradictory hypothesis that $\bar{c}_k > 0$, along with $\theta > 0$, we have

$$\langle \mathbf{c}, \mathbf{x}^* + \theta \mathbf{d}^{(k)} \rangle = \langle \mathbf{c}, \mathbf{x}^* \rangle + \theta \bar{c}_k > \langle \mathbf{c}, \mathbf{x}^* \rangle,$$

which contradicts the hypothesis that $\mathbf{x}^*$ attains the maximum. $\square$

Under Linear Program (LPS) in Assumption 4.7, by definition, we have the choice $\mathbf{x} \in P$ is a basic feasible solution implies that $M_B[\mathbf{x}]_B = \mathbf{b}$ and $[\mathbf{x}]_B \geq 0$. Now, since M_B is invertible, we have that $[\mathbf{x}]_B = M_B^{-1}\mathbf{b}$; the two parts in the necessary condition can be combined into $M_B^{-1}\mathbf{b} \geq 0$. Meanwhile, observe that the specification in the sufficient condition of Theorem 4.12 is to specify a basis since the assumption on degeneracy is missing; thus, if we include it, we may solely discuss the basis submatrix of the constraint matrix, independent of a specific point in P . As such, consider the following definition.

Definition 4.18. Consider Linear Program (LPS) in Assumption 4.7. The matrix M_B is *optimal* if it satisfies that $M_B^{-1}\mathbf{b} \geq 0$ and the reduced cost $\bar{\mathbf{c}} = \mathbf{c} - (M_B^{-1}M)^\top[\mathbf{c}]_B \leq 0$.

4.4 Moving Between Basic Feasible Solutions

As an algorithm, the simplex method takes in a linear program. The algorithm also optionally takes in a basic feasible solution to initiate the procedure or, if none is given, proceeds with a preliminary procedure to generate one; we will discuss this procedure in the next section. Thus, we study the so-called *Phase II* of the simplex method in this section; in particular, Phase II also provides intuition for which basic feasible solution to use if it is not provided.

Based on Linear Program (LPS) in Assumption 4.7, we first assume there is a basic feasible solution $\mathbf{x}$ that is nondegenerate and define the unique B and induce the invertible matrix M_B . Compute each k -th reduced cost with respect to basis B and form $\mathbf{c}$; by Theorem 4.12, if $\bar{\mathbf{c}} \leq 0$, then $\mathbf{x}$ attains the maximum of $\langle \mathbf{c}, \cdot \rangle$ over P .

We now suppose that there is some k so that $\bar{c}_k > 0$, which repeats exactly the proof of Theorem 4.17, and we know that there is some $\theta \in \mathbb{R}_{++}$ so that $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$; in particular, Theorem 4.17 relies on Lemma 4.15 for the choice of θ . We now show that this choice leads to a basic feasible solution with respect to another basis.

Theorem 4.19. Consider Linear Program (LPS) in Assumption 4.7 and let $\mathbf{x}$ be a basic feasible solution so that $\text{supp}(\mathbf{x}) \subseteq B$. Suppose there is $k \in N$ such that $\overline{c}_k > 0$; either (LPS) is not bounded from above over P , or there is some $i^* \in B$ where

$$\theta = -\frac{[\mathbf{x}]_{i^*}}{[\mathbf{d}^{(k)}]_{i^*}} = \min_{i \in B, [\mathbf{d}^{(k)}]_i < 0} -\frac{[\mathbf{x}]_i}{[\mathbf{d}^{(k)}]_i}$$

that defines another basic feasible solution $\mathbf{x}' = \mathbf{x} + \theta \mathbf{d}^{(k)} \in P$. In particular, if $\mathbf{x}$ is nondegenerate, then $\theta > 0$.

Proof: By Theorem 4.17, $\mathbf{x}$ is not optimal. By the choice of $\mathbf{x}'$ in Lemma 4.15 and the constructive proof, if $\mathbf{d}^{(k)} \geq \mathbf{0}$, then the optimization problem $\max_{\mathbf{x} \in P} \langle \mathbf{c}, \mathbf{x} \rangle$ is not bounded from above. In the following, we may assume that there is some $i \in N$ so that $[\mathbf{d}^{(k)}]_i < 0$.

Define θ and $i^* \in B$ as in the statement and that $\mathbf{x}' = \mathbf{x} + \theta \mathbf{d}^{(k)}$; observe that

$$[\mathbf{x}]_{i^*} - \frac{[\mathbf{x}]_{i^*}}{[\mathbf{d}^{(k)}]_{i^*}} [\mathbf{d}^{(k)}]_{i^*} = 0.$$

We now need to further investigate the entries of $\mathbf{x}$ and $\mathbf{x}'$; without loss of generality, we may assume $i^* = 1$, $B = \{1, 2, \dots, m\}$, $k = m + 1$, and $N = \{m + 1, m + 2, \dots, n\}$. Consider the construction of Lemma 4.15, we write

$$\mathbf{x}' = \begin{bmatrix} [\mathbf{x}]_B \\ 0 \\ \mathbf{0} \end{bmatrix} + \theta \begin{bmatrix} [\mathbf{d}^{(k)}]_B \\ 1 \\ \mathbf{0} \end{bmatrix}$$

It is now clear that we have designed to have $\text{supp}(\mathbf{x}') \setminus \{k\} \subseteq B \setminus \{i^*\}$ since the choice of i^* may not be unique but $[\mathbf{x}']_{i^*} = 0$ necessarily. Let $B' = B \setminus \{i^*\} \cup \{k\}$; we will show that $B' = \{2, 3, \dots, m, m + 1\}$ corresponds to a list of linearly independent columns of M . Let $(\mu_j)_{j \in B'}$ be a list of real numbers so that $\mathbf{0} = \sum_{j \in B'} \mu_j \mathbf{f}_j$; we apply M_B^{-1} to both sides of elements in $\mathbb{R}^m$ so that

$$\begin{aligned} \mathbf{0} = M_B^{-1} \mathbf{0} &= M_B^{-1} (\mu_2 \mathbf{f}_2 + \mu_3 \mathbf{f}_3 + \dots + \mu_{m+1} \mathbf{f}_{m+1}) \\ &= \mu_2 M_B^{-1} \mathbf{f}_2 + \mu_3 M_B^{-1} \mathbf{f}_3 + \dots + \mu_{m+1} M_B^{-1} \mathbf{f}_{m+1} \\ &= \mu_2 \boldsymbol{\chi}_2 + \mu_3 \boldsymbol{\chi}_3 + \dots + \mu_m \boldsymbol{\chi}_m + \mu_{m+1} [\mathbf{d}^{(k)}]_B \end{aligned} \tag{4.20}$$

where the first $m - 1$ terms are due to the observation that $M_B^{-1} \mathbf{f}_j = \boldsymbol{\chi}_j$ for $j \in B$ and the last term is given by the choice of k gives that $M_B^{-1} \mathbf{f}_k = -[\mathbf{d}^{(k)}]_B$. Since we have assumed $[\mathbf{d}^{(k)}]_{i^*} < 0$, where $i^* = 1$ and, in (4.20), all $\boldsymbol{\chi}_i$ for $i \in \{2, 3, \dots, m\}$ does not contribute to the first entry, we may conclude that $\mu_{m+1} = 0$. Now, since the list $\boldsymbol{\chi}_2, \boldsymbol{\chi}_3, \dots, \boldsymbol{\chi}_m$ is linearly independent, combined with $\mathbf{0} = 0[\mathbf{d}^{(k)}]_B$, we have that

$\{\mu_2, \mu_3, \dots, \mu_m\} = \{0\}$. This concludes that $\chi_2, \chi_3, \dots, \chi_m, [\mathbf{d}^{(k)}]_B$ is a list of linearly independent vectors and, by properties of invertible mappings, $\mathbf{f}_2, \mathbf{f}_3, \dots, \mathbf{f}_{m+1}$ is a list of linearly independent vectors. By Theorem 3.23 and Theorem 4.4, $\mathbf{x}'$ is a basic feasible solution. $\square$

We remark a few points from the proof of Theorem 4.19. First, observe that it provides an inductive step for implementing the simplex algorithm. Consider a basic index set B associated with a basic feasible solution $\mathbf{x}$ such that $\text{supp}(\mathbf{x}) \subseteq B$. Theorem 4.19 computes a new basic feasible solution and updates to a new basic index set $B' = B \setminus \{k\} \cup \{i^*\}$. We will call an index, a column, or a variable *exiting* if it is associated with k and call them *entering* if they are associated with i^* . We may refer to the addition $\mathbf{x} + \theta \mathbf{d}^{(k)}$ by *moving away* from $\mathbf{x}$ or *entering* $\mathbf{x} + \theta \mathbf{d}^{(k)}$ if $\theta > 0$. A strategy of choosing such an i^* , regardless of whether it is unique or not, is called a *pivoting rule*.

However, observe that the choice of i^* does not guarantee that $\theta \neq 0$ in a degenerate case; this, though, does not break the algorithm but may cause an infinite loop so that the algorithm never terminates. Thus, it forces us to introduce a pivoting rule that never enters an existing basis. We will show in the next section that there is one strategy. Given a nondegenerate basic feasible solution $\mathbf{x}$ and the associated basis B , we build the following inductive step.

Definition 4.21 (The Simplex Method). Consider Linear Program (LPS) in Assumption 4.7 and a given pivoting rule.

1. Compute

$$\bar{\mathbf{c}} = \mathbf{c} - (M_B^{-1}M)^\top [\mathbf{c}]_B$$

and determine whether or not $\bar{\mathbf{c}} \leq \mathbf{0}$; if so, then return $\mathbf{x}$ as the optimal solution to the linear program; otherwise, let $k \in N$ be an index so that $\bar{c}_k > 0$.

2. Compute $\mathbf{d}^{(k)}$. If $\mathbf{d}^{(k)} \geq \mathbf{0}$, then terminate the procedure and inform that the linear program is unbounded from above. Otherwise, update the following two quantities:
3. Choose the scalar θ and an associated index i^* under the pivoting rule so that

$$\theta = -\frac{[\mathbf{x}]_{i^*}}{[\mathbf{d}^{(k)}]_{i^*}} = \min_{i \in B, [\mathbf{d}^{(k)}]_i < 0} -\frac{[\mathbf{x}]_i}{[\mathbf{d}^{(k)}]_i}$$

and compute $\mathbf{x}' = \mathbf{x} + \theta \mathbf{d}^{(k)}$.

4. Update $B' = B \setminus \{i^*\} \cup \{k\}$.

4.5 Lexicographical Order

For two distinct vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we say $\mathbf{u}$ is *lexicographically smaller than* $\mathbf{v}$ and denote by $\mathbf{u} \prec \mathbf{v}$ if the first nonzero component of $\mathbf{v} - \mathbf{u}$ is positive; in other words, the first distinct entry u_i and v_i has that $u_i < v_i$. Respectively, $\mathbf{u}$ is lexicographically larger than $\mathbf{v}$ if the first nonzero component of $\mathbf{v} - \mathbf{u}$ is negative. We abbreviate the comparison as *lexicolarger* or *lexicosmaller*. This notation extends to the comparison with $\mathbf{0}$: a point $\mathbf{u}$ is *lexicopositive* if $\mathbf{0} \prec \mathbf{u}$ and is *lexiconegative* if $\mathbf{0} \prec -\mathbf{u}$.

Lemma 4.22. *The lexicographical ordering is a strict total order.*

Proof: Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and u_i, v_i, w_i denote the i -th entry $\mathbf{u}, \mathbf{v}, \mathbf{w}$, respectively. First, irreflexivity holds because it is not defined for a pair of identical vectors. If $\mathbf{u} \prec \mathbf{v}$ for two vectors, then the first index i such that $u_i < v_i$ and $\mathbf{v} \not\prec \mathbf{u}$; thus, lexicographical ordering is asymmetric. Suppose $\mathbf{u} \prec \mathbf{v}$ and $\mathbf{v} \prec \mathbf{w}$. There are indices k and ℓ so that $u_k < v_k$, $u_i = v_i$ for $i \in \{1, 2, \dots, k\}$, $v_\ell < w_\ell$, and $v_j = w_j$ for $j \in \{1, 2, \dots, \ell\}$. If $k \leq \ell$, then $u_k < v_k = w_k$; thus, $\mathbf{u} \prec \mathbf{w}$. If $\ell < k$, then $u_\ell = v_\ell < w_\ell$; thus, $\mathbf{u} \prec \mathbf{w}$ again. Therefore, the transitivity holds. The connectedness holds naturally. $\square$

Consider Linear Program (LPS) in Assumption 4.7 and $\mathbf{x}$ is basic feasible solution that there is $k \in N$ so that $\bar{c}_k > 0$. Let $\mathbf{d}^{(k)} = M_B^{-1} \mathbf{f}_k$ be the basic direction. As $\bar{c}_k > 0$, $\mathbf{x}$ is not optimal and, by Corollary 4.16, the objective function is unbounded if $\mathbf{d}^{(k)} \geq \mathbf{0}$. We may assume that there are one or more entries of $\mathbf{d}^{(k)}$ that are negative.

In an engineering and implementation perspective, we consider the following augmented matrix. Consider the matrix obtained by attaching $\mathbf{b}$ as the first column of M and apply the transformation

$$M_B^{-1}[\mathbf{b} \mid M] = M_B^{-1}[\mathbf{b} \mid \mathbf{f}_1 \mid \mathbf{f}_2 \mid \cdots \mid \mathbf{f}_n].$$

As we have investigated in Section 4.3, the resulting matrix consists of the following elements.

- $M_B^{-1}\mathbf{b} = M_B^{-1}M\mathbf{x}$ determines whether the solution is optimal; if not, we need the following vectors.
- Each column $M_B^{-1}\mathbf{f}_k$ either is a basic direction $-\mathbf{d}^{(k)}$ if $k \in N$ or an standard basis vector $\mathbf{x}_j$ for $k = B(j)$ for $k \in B$.

We now also integrate the reduced cost into consideration. Recall the following facts. The reduced cost $\bar{\mathbf{c}}$ can be written in a vectorized form that $\bar{\mathbf{c}}^\top = \mathbf{c}^\top - [\mathbf{c}]_B^\top M_B^{-1}M$. Theorem 3.23 addressed that, if $\mathbf{x}$ is a basic feasible solution, then $[\mathbf{x}]_N = \mathbf{0}$ is the zero vector; thus, a basic feasible solution $\mathbf{x}$ has that $\langle \mathbf{c}, \mathbf{x} \rangle = \langle [\mathbf{c}]_B, [\mathbf{x}]_B \rangle$ is the objective value.

As $M_B[\mathbf{x}]_B = \mathbf{b}$, we have that $M_B[\mathbf{x}] = \mathbf{b}$. Consider the $(m+1) \times (n+1)$ -matrix

$$\left[\begin{array}{c|c} -\langle [\mathbf{c}]_B, M_B^{-1}\mathbf{b} \rangle & \mathbf{c}^\top - [\mathbf{c}]_B^\top M_B^{-1}M \\ \hline M_B^{-1}\mathbf{b} & M_B^{-1}M \end{array} \right]$$

and the entries of the matrix can be expanded into

$$\left[\begin{array}{c|cccc} -\langle \mathbf{c}, \mathbf{x} \rangle & \overline{c}_1 & \overline{c}_2 & \dots\dots & \overline{c}_n \\ \hline [\mathbf{x}]_{B(1)} & \vdots & \vdots & & \vdots \\ [\mathbf{x}]_{B(2)} & | & | & & | \\ \vdots & -\mathbf{d}^{(1)} & -\mathbf{d}^{(2)} & \dots\dots & -\mathbf{d}^{(n)} \\ \vdots & | & | & & | \\ [\mathbf{x}]_{B(m)} & \vdots & \vdots & & \vdots \end{array} \right].$$

We call this matrix the *tableau* or *simplex tableau* of Linear Program (LPS) defined by the basis B , the *tableau* defined by the basis B , or simply the *tableau* given the clarity of the context. In a given tableau, we index the rows and columns and name a part of it as follows.

- The leading column that corresponds to a basic feasible solution as the *zero-th column* so that the numbering of the remaining columns is consistent with the numbering in M itself;
- The leading row that corresponds to the current objective value and the reduced cost as the *zero-th row*.
- We call the part of the tableau consisting of all rows except the zero-th row $M_B^{-1}[\mathbf{b} \mid M]$ the *body* of the tableau.

Let T be a tableau; we denote the i -th row of T by T_i for $i \in \{0, 1, 2, \dots, m\}$, the j -th column as $T_{:,j}$ for $j \in \{0, 1, 2, \dots, n\}$, and the (i, j) -th entry as $T_{i,j}$.

Be aware that in different literature, the simplex tableau is also known as the full tableau and the body of a tableau as the simplex tableau. We consider the following choosing strategy for the entry criterion. As we mentioned in Remark 4.8, the index set B is now a list and its order matters.

Definition 4.23 (Lexicographic Pivoting Rule). Given that T is a tableau defined by a basis B . Let $k \in N$ such that $\overline{c}_k > 0$ and obtain $\mathbf{d}^{(k)} \in \mathbb{R}^m$. Let $I = \{i \in B : [\mathbf{d}^{(k)}]_i < 0\}$. Consider a second tableau obtained by dividing each i -th row, for $i \in I$, of T by $-[\mathbf{d}^{(k)}]_i$. Let $\ell \in \{1, 2, \dots, m\}$ be the lexicographically smallest row in the

second tableau and let i^* be the index such that $B(\ell) = i^*$; define B' by replacing $B(\ell)$ with k .

In the rule, the index set I is nonempty; if $I = \emptyset$, then the algorithm terminates because the problem is unbounded. Note that the lexicographic pivoting rule does not modify the tableau; it makes a copy to compute the comparison. Also note that the row selection seems not to involve the 0-th row but, in fact, $\mathbf{d}^{(k)}$ is already embedded into the tableau when performing the division. Notice that the lexicographic pivoting rule does not explicitly compute the stepsize θ , but the resulting 0-th column embeds the nonzero part of the basic feasible solution. Last but not least, note that the body of the tableau does not involve the optimization problem while the 0th row does.

Theorem 4.24. *Consider Linear Program (LPS) in Assumption 4.7 and a basic feasible solution $\mathbf{x}$ to it; we further suppose that every row except the 0-th row of its tableau is lexicographically positive. Suppose the algorithm proceeds with the lexicographic pivoting rule; the following statement holds.*

1. *Every row, except the 0-th row, of the tableau inductively remains lexi-positive.*
2. *Inductively, the 0-th row strictly decreases lexicographically.*
3. *The simplex method algorithm terminates after a finite number of iterations.*

Proof: Suppose we start with a tableau in which all rows are lexicographically positive except the 0th row. Let $k \in N$ so that $\bar{c}_k > 0$; we denote $\mathbf{u} = -\mathbf{d}^{(k)} = M_B^{-1} \mathbf{f}_k$ and denote each entry by $u_\ell = -[\mathbf{d}^{(k)}]_\ell$ for each $\ell \in \{1, 2, \dots, m\}$. Let $I = \{i \in B : u_i > 0\}$ and S be the tableau obtained by the masked division with the lexicographical pivoting rule on each row with index in I and ℓ be the index that satisfies

$$\frac{1}{u_\ell} T_\ell \prec \frac{1}{u_j} T_j \quad \text{for all } j \in \{1, 2, \dots, m\} \setminus \{\ell\}$$

and obtain a new index set B' by $B'(j) = B(j)$ for all $j \in \{1, 2, \dots, m\} \setminus \{\ell\}$ and $B'(\ell) = k$. We now have a new tableau T' defined by the basis B' . Note that in both tableaux T and T' , the columns $T_{:,B(\ell)}$ and $T'_{:,B'(\ell)}$ are both the standard basis vector $\mathbf{x}_\ell \in \mathbb{R}^m$; that is,

$$M_B^{-1} \mathbf{f}_{B(\ell)} = M_{B'}^{-1} \mathbf{f}_{B'(\ell)} = \mathbf{x}_\ell.$$

We now define $E \in \mathbb{R}^{m \times m}$ as the column exchange matrix in the form of

$$E = [\mathbf{x}_1 \mid \mathbf{x}_2 \mid \dots \mid \mathbf{x}_{\ell-1} \mid \mathbf{u} \mid \mathbf{x}_{\ell+1} \mid \dots \mid \mathbf{x}_m]$$

in which we have that

$$M_B E = [\mathbf{f}_{B(1)} \mid \mathbf{f}_{B(2)} \mid \dots \mid M_B \mathbf{f}_{B(\ell-1)} \mid \mathbf{u} \mid \mathbf{f}_{B(\ell+1)} \mid \dots \mid \mathbf{f}_{B(m)}]$$

in which the ℓ -th column can be rewritten as $M_B M_B^{-1} \mathbf{f}_k = \mathbf{f}_k$, which was by the definition of $\mathbf{u}$. Yet, observe that $M_B E$ now follows precisely the definition of $M_{B'}$ since $k = B'(\ell)$ and $B(j) = B'(j)$ if $j \in \{1, 2, \dots, m\} \setminus \{\ell\}$.

Define the matrix $Q \in \mathbb{R}^{m \times m}$ with each row as

$$Q_\ell = \frac{1}{u_\ell} \boldsymbol{\chi}_\ell^\top \quad \text{and} \quad Q_i = \boldsymbol{\chi}_i^\top - \frac{u_i}{u_\ell} \boldsymbol{\chi}_\ell^\top \text{ for } i \in \{1, 2, \dots, m\} \setminus \ell.$$

We show that Q is the inverse of E ; that is, $QE = I_m$. Observe that at the ℓ -th column of QE is $Q\mathbf{u}$ and we have that

$$[Q\mathbf{u}]_\ell = \langle Q_\ell^\top, \mathbf{u} \rangle = \left\langle \frac{1}{u_\ell} \boldsymbol{\chi}_\ell, \mathbf{u} \right\rangle = \frac{1}{u_\ell} \langle \boldsymbol{\chi}_\ell, \mathbf{u} \rangle = \frac{1}{u_\ell} u_\ell = 1$$

and

$$[Q\mathbf{u}]_i = \langle Q_i^\top, \mathbf{u} \rangle = \left\langle \boldsymbol{\chi}_i - \frac{u_i}{u_\ell} \boldsymbol{\chi}_\ell, \mathbf{u} \right\rangle = \langle \boldsymbol{\chi}_i, \mathbf{u} \rangle - \frac{u_j}{u_\ell} \langle \boldsymbol{\chi}_\ell, \mathbf{u} \rangle = u_i - u_i = 0.$$

Let $i \in \{1, 2, \dots, m\} \setminus \{\ell\}$; for j -th column of QE , for $j \in \{1, 2, \dots, m\} \setminus \{\ell\}$, we have $Q\boldsymbol{\chi}_j = Q_{:,j}$; this is straightforward to verify that

$$[Q\boldsymbol{\chi}_j]_\ell = Q_{\ell,j} = \left[\frac{1}{u_\ell} \boldsymbol{\chi}_\ell^\top \right]_j = 0$$

and

$$[Q\boldsymbol{\chi}_j]_i = Q_{i,j} = \left[\boldsymbol{\chi}_i - \frac{u_i}{u_\ell} \boldsymbol{\chi}_\ell \right]_j = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases}$$

This now verifies that Q is indeed the inverse of E . Observe that, as $M_B E = M_{B'}$, we have that $M_{B'}^{-1} = (M_B E)^{-1} = E^{-1} M_B^{-1} = Q M_B^{-1}$; further, we may write

$$T' = M_{B'}^{-1} [\mathbf{b} \mid M] = Q M_B^{-1} [\mathbf{b} \mid M] = QT.$$

This provides us with a method for comparing the two tableaux using linear algebra; observe that

$$T'_\ell = \frac{1}{u_\ell} T_\ell \quad \text{and} \quad T'_j = T_j - \frac{u_j}{u_\ell} T_\ell \quad \text{for } j \in \{1, 2, \dots, m\} \setminus \{\ell\}.$$

By the definition of ℓ , we have that $u_\ell = [-\mathbf{d}^{(k)}]_\ell > 0$, which first gives the lexicographical positiveness of T'_ℓ . Let $j \in \{1, 2, \dots, m\} \setminus \{\ell\}$. If $u_j > 0$, we have chose $\frac{1}{u_\ell} T_\ell \prec \frac{1}{u_j} T_j$ and rearrange and multiply by $u_j > 0$, we have $\mathbf{0} \prec T_j - \frac{u_j}{u_\ell} T_\ell$; by definition, we have that T'_j is lexicographically positive. Now, if $u_j \leq 0$, then $-u_j/u_\ell \geq 0$; coupled with the fact that all rows are lexicographically positive, we have that $T'_j = T_j + (-u_j/u_\ell) T_\ell$ is lexicographically positive. $\square$